

Setu Reconciliation API Documentation

Developer API Reference & Integration Guide

Backend Service: FastAPI

Database: PostgreSQL

Date: July 2026

Version: 1.0.0

Overview

This service provides payment lifecycle event ingestion and reconciliation capabilities. It supports event logging with strict idempotency check, transaction state maintenance, and discrepancy checks for reporting. This document describes the schema structure, request payloads, query parameters, responses, and curl commands for all active API endpoints.

Base URL

Production Server: <http://43.205.94.73:8000>

1. Ingest Events

[POST] /events

Accepts payment lifecycle events (payment_initiated, payment_processed, payment_failed, settled). Event ingestion is idempotent, duplicate events (same event_id) return success without mutating transaction history.

Request Payload (JSON)

```
{
  "event_id": "b768e3a7-9eb3-4603-b21c-a54cc95661bc",
  "event_type": "payment_initiated",
  "transaction_id": "2f86e94c-239c-4302-9874-75f28e3474ee",
  "merchant_id": "merchant_2",
  "merchant_name": "FreshBasket",
  "amount": 15248.29,
  "currency": "INR",
  "timestamp": "2026-01-08T12:11:58.085567+00:00"
}
```

Response Payload (JSON)

```
{
  "event_id": "b768e3a7-9eb3-4603-b21c-a54cc95661bc",
  "event_type": "payment_initiated",
  "amount": "15248.29",
  "currency": "INR",
  "timestamp": "2026-01-08T12:11:58.085567+00:00",
  "received_at": "2026-07-19T17:37:46.429153+05:30"
}
```

Sample curl Command

```
curl -X POST http://43.205.94.73:8000/events \
-H "Content-Type: application/json" \
-d '{
  "event_id": "b768e3a7-9eb3-4603-b21c-a54cc95661bc",
  "event_type": "payment_initiated",
  "transaction_id": "2f86e94c-239c-4302-9874-75f28e3474ee",
  "merchant_id": "merchant_2",
  "merchant_name": "FreshBasket",
  "amount": 15248.29,
  "currency": "INR",
  "timestamp": "2026-01-08T12:11:58.085567+00:00"
}'
```

2. List Transactions

[GET] /transactions

Retrieves transactions with support for advanced filtering, sorting, pagination, and date-range conditions.

Query Parameters

Parameter	Type	Req	Description
merchant_id	string	No	Filter by Merchant ID
status	string	No	Filter by status (payment_initiated, payment_processed, payment_failed, settled)
start_date	datetime	No	Filter created on or after ISO timestamp
end_date	datetime	No	Filter created on or before ISO timestamp
skip	integer	No	Offset for pagination (default: 0)
limit	integer	No	Limit for pagination (default: 100)
sort_by	string	No	Sort by field: created_at, updated_at, amount, status (default: created_at)
sort_order	string	No	Sort direction: asc, desc (default: desc)

Response Payload (JSON)

```
[
  {
    "id": "2f86e94c-239c-4302-9874-75f28e3474ee",
    "merchant_id": "merchant_2",
    "merchant_name": "FreshBasket",
    "amount": "15248.29",
    "currency": "INR",
    "status": "payment_failed",
    "has_discrepancy": false,
    "discrepancy_reason": null,
    "created_at": "2026-01-08T12:11:58.085567+05:30",
    "updated_at": "2026-01-08T12:38:58.085567+05:30"
  }
]
```

Sample curl Command

```
curl -X GET "http://43.205.94.73:8000/transactions?merchant_id=merchant_2&status=payment_failed&limit=2"
```

3. Fetch Transaction Details

[GET] /transactions/{transaction_id}

Returns complete details for a single transaction, including its associated merchant data and its full chronological event history/state transitions.

Query Parameters

Parameter	Type	Req	Description
transaction_id	string	Yes	Path Parameter. The unique transaction identifier.

Response Payload (JSON)

```
{
  "id": "2f86e94c-239c-4302-9874-75f28e3474ee",
  "merchant_id": "merchant_2",
  "merchant_name": "FreshBasket",
  "amount": "15248.29",
  "currency": "INR",
  "status": "payment_failed",
  "has_discrepancy": false,
  "discrepancy_reason": null,
  "created_at": "2026-01-08T12:11:58.085567+05:30",
  "updated_at": "2026-01-08T12:38:58.085567+05:30",
  "events": [
    {
      "event_id": "b768e3a7-9eb3-4603-b21c-a54cc95661bc",
      "event_type": "payment_initiated",
      "amount": "15248.29",
      "currency": "INR",
      "timestamp": "2026-01-08T12:11:58.085567+05:30",
      "received_at": "2026-07-19T17:37:46.429153+05:30"
    },
    {
      "event_id": "da46895f-4b47-4505-900e-d067f64a55eb",
      "event_type": "payment_failed",
      "amount": "15248.29",
      "currency": "INR",
      "timestamp": "2026-01-08T12:38:58.085567+05:30",
      "received_at": "2026-07-19T17:37:46.574057+05:30"
    }
  ]
}
```

Sample curl Command

```
curl -X GET http://43.205.94.73:8000/transactions/2f86e94c-239c-4302-9874-75f28e3474ee
```

4. Reconciliation Summary

[GET] /reconciliation/summary

Returns aggregated transaction volume, value, and discrepancy counts grouped by dimensions: merchant, calendar date, and state status.

Response Payload (JSON)

```
{
  "by_merchant": [
    {
      "dimension": "merchant",
      "group_value": "FreshBasket (merchant_2)",
      "total_transactions": 810,
      "total_amount": "21040319.45",
      "discrepancy_count": 52
    }
  ],
  "by_date": [
    {
      "dimension": "date",
      "group_value": "2026-01-08",
      "total_transactions": 250,
      "total_amount": "5204810.15",
      "discrepancy_count": 12
    }
  ],
  "by_status": [
    {
      "dimension": "status",
      "group_value": "settled",
      "total_transactions": 2341,
      "total_amount": "61023049.20",
      "discrepancy_count": 48
    }
  ]
}
```

Sample curl Command

```
curl -X GET http://43.205.94.73:8000/reconciliation/summary
```

5. Reconciliation Discrepancies

[GET] /reconciliation/discrepancies

Returns transactions flagged with status/amount inconsistencies. Discrepancies include: settlements on failed payments, processed events but never settled (>6 hrs), duplicate conflicting events, and amount/currency mismatches.

Query Parameters

Parameter	Type	Req	Description
skip	integer	No	Offset for pagination (default: 0)
limit	integer	No	Limit for pagination (default: 100)

Response Payload (JSON)

```
[
  {
    "id": "02d878de-f807-4bf1-9b16-7098be6e54fe",
    "merchant_id": "merchant_5",
    "merchant_name": "StyleHub",
    "amount": "41263.87",
    "currency": "INR",
    "status": "settled",
    "has_discrepancy": true,
    "discrepancy_reason": "Duplicate conflicting events of type 'settled'",
    "created_at": "2026-01-08T14:29:10.700477+05:30",
    "updated_at": "2026-01-08T22:10:10.700477+05:30",
    "events": []
  }
]
```

Sample curl Command

```
curl -X GET "http://43.205.94.73:8000/reconciliation/discrepancies?limit=2"
```